

ECON 636 600: MACROECONOMIC THEORY I

Assignment 5 (due on 10:00am Tuesday, Nov 13)

1. (60 points) Numerically solve the following problem. Consider an economy where an agent can be in one of three states:

- (a) e : the agent is employed and his income is y .
- (b) u : the agent is unemployed and is eligible for unemployment benefits b ($b < y$).
- (c) $\tilde{u}$: the agent is unemployed and is not eligible for unemployment benefits (i.e., his benefits expired). He receives welfare payment d ($d < b$).

If a is the agent's asset holdings, let $V_i(a)$ be the agent's value function if his state is $i \in \{e, u, \tilde{u}\}$. $V_e(a)$ satisfies

$$V_e(a) = \max_{a'} \frac{(c^{1-\rho}(1-h)^\rho)^{1-\sigma}}{1-\sigma} + \beta[\pi_{eu}V_u(a') + (1-\pi_{eu})V_e(a')]$$

where π_{eu} is the job-loss probability, $c = (1+r)a + y - a'$ is consumption, h is labor supply.

$V_u(a)$ satisfies

$$V_u(a) = \max_{a'} \frac{(c^{1-\rho})^{1-\sigma}}{1-\sigma} + \beta[\pi_{ue}V_e(a') + \pi V_{\tilde{u}}(a') + (1-\pi_{ue}-\pi)V_u(a')],$$

where π_{ue} is the job-finding probability and π is the probability that unemployment benefits expire, $c = (1+r)a + b - a'$ is consumption. $V_{\tilde{u}}(a)$ satisfies

$$V_{\tilde{u}}(a) = \max_{a'} \frac{(c^{1-\rho})^{1-\sigma}}{1-\sigma} + \beta[\pi_{ue}V_e(a') + (1-\pi_{ue})V_{\tilde{u}}(a')],$$

where $c = (1 + r)a + d - a'$ is consumption. The parameters are

$$\beta = 0.995$$

$$\rho = 0.67$$

$$\sigma = 2.5$$

$$h = 0.45$$

$$r = 0.0025$$

$$y = 1$$

$$b = 0.5$$

$$d = 0.1$$

$$\pi_{ue} = 0.5$$

$$\pi_{eu} = 0.0319$$

$$\pi = 0.25$$

$$a, a' \in \{0, 0.01, 0.02, \dots, 8\}$$

Compute the policy functions (savings as a function of asset) for each type of agent by grid search. Plot the policy functions.

Fortran code and the output file “Assignment5_Q1.csv” are in elearning.

2. An economy consists of two infinitely lived consumers named $i = 1, 2$. There is one nonstorable consumption good. Consumer i consumes c_t^i at time t . Consumer i ranks consumption streams by

$$\sum_{t=0}^{\infty} \beta^t u(c_t^i),$$

where $\beta \in (0, 1)$ and $u(c)$ is increasing, strictly concave, and twice continuously differentiable.

Consumer 1 is endowed with a stream of the consumption good $y_t^1 = 1, 0, 0, 1, 0, 0, 1, \dots$

Consumer 2 is endowed with a stream of the consumption good $y_t^2 = 0, 1, 1, 0, 1, 1, 0, \dots$

Assume that there are complete markets with time-0 trading.

(a) Define a competitive equilibrium.

(b) Compute a competitive equilibrium.

consumptions are perfectly smoothed. Hence $q_t(s^t) = \beta^t \pi(s^t)$. consumption $\bar{c}^1 = (1 - \beta)(1 + \beta^3 + \beta^6 + \dots) = \frac{1-\beta}{1-\beta^3}$. consumption $\bar{c}^2 = (1 - \beta)(\beta^1 + \beta^2 + \beta^4 + \beta^5 + \dots) = \frac{(1-\beta)(\beta^1 + \beta^2)}{1-\beta^3}$.

(c) Suppose that one of the consumers markets a derivative asset that promises to pay .05 units of consumption each period. What would the price of that asset be?

$$\sum_t \sum_{s^t} \beta^t \pi(s^t) 0.05 = \sum_t \beta^t 0.05 = \frac{0.05}{1-\beta}$$